

D638

Monitoring Student Learning Through Assessment

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A teacher in a school with a new greenhouse gives a 10-question quiz midway through a unit, uses results to regroup students for reteaching the next day, and records no grade. The quiz is printed on blue paper. How is the quiz being used?

- A. Formatively.
- B. As a benchmark assessment.
- C. Summatively, because it is a quiz.
- D. As a norm-referenced test.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A parent at a school with a strong music program reads that her son scored at the 70th percentile on a national reading test. She has three children in the district. She concludes he got 70% of the questions right. How should the teacher explain the score?

- A. He scored better than his two siblings.
- B. He scored as well as or better than about 70% of students in the comparison group.
- C. He is 70% proficient on the standards.
- D. He answered 70% of questions correctly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A state test reports each student as below, approaching, meeting, or exceeding the standard based on fixed cut scores. What type of interpretation is this?

- A. Criterion-referenced.
- B. Formative.
- C. Norm-referenced.
- D. Percentile-based.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A math test on solving word problems is written at a reading level three grades above the students' level. Students' scores are nearly identical when they retake it a week later. The test was designed by a team of five teachers. What is the main concern?

- A. Reliability: scores are inconsistent.
- B. Validity: scores may reflect reading ability rather than math problem solving.
- C. None; consistent scores prove the test is valid.
- D. The test needs a larger design team.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A second grader in a school with a new reading coach has weekly fluency probes. The aim line calls for gains of 2 words correct per minute each week. Her last four scores fell below the aim line. She loves art class. What should the team do?

- A. Add art to the intervention.
- B. Raise the goal.
- C. Keep the intervention; wait for more data.
- D. Change the intervention, since several consecutive points are below the aim line.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A student read 40 words correct per minute in week 1 and 64 in week 9. What is the average weekly rate of improvement?

- A. 24 words per week
- B. 7.1 words per week
- C. 3 words per week
- D. 2.67 words per week

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

On a unit test given in a newly remodeled classroom, 30 students answered item 7. Only 9 answered correctly, and most of the top-scoring students chose the same wrong answer. The test took 45 minutes. What should the teacher conclude?

- A. The test should be shorter than 45 minutes.
- B. The item is fine; top students should have studied more.
- C. The item is hard (0.30) and likely flawed; review it for a miskeyed or confusing answer.
- D. The item is easy, since its difficulty value is low.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

An item has a difficulty of 0.85 on a test taken by 40 students. How many students answered it correctly?

- A. 85
- B. 47
- C. 6
- D. 34

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

A teacher at a school with a new theater wants students to see exactly which parts of their argumentative essays need work: claim, evidence, reasoning, and conventions. She teaches 110 students. Which scoring tool fits best?

- A. An analytic rubric with a separate score for each criterion.
- B. A single letter grade.
- C. A holistic rubric, since she has many students.
- D. A checklist of whether the essay was turned in.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A student in a school with a large library turns in a lab report. The teacher, who coaches volleyball, returns it three weeks later. Which feedback would be most effective if given promptly?

- A.** Great effort, like in volleyball!
- B.** B+. Good job.
- C.** Needs improvement throughout.
- D.** Your conclusion states the result but not whether it supports your hypothesis; add one sentence linking them.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher at a school with a new front office is presenting test results to families. Scores for ten students were 72, 75, 76, 78, 78, 80, 81, 82, 84, and 4 (a student who left after one question). Which statistic best represents the typical score?

- A.** The range, 80.
- B.** The lowest score, 4.
- C.** The median, 78.
- D.** The mean, 71.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A class's test results show a mean of 74 and a median of 86. What does this pattern most likely mean?

- A.** Most students scored about 74.
- B.** A few very low scores are pulling the mean down; most students scored in the 80s.
- C.** One of the statistics must be miscalculated, since they should always match.
- D.** A few very high scores are pulling the mean up.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A teacher ~~in a school with a new greenhouse~~ gives a 10-question quiz midway through a unit, uses results to regroup students for reteaching the next day, and records no grade. ~~The quiz is printed on blue paper.~~ How is the quiz being used?

Target word: classify by use **Decides:** Results adjust instruction; no grade recorded.

A	Correct	Results guide next steps in teaching.
B	Adjacent category	Benchmarks are periodic, common checks across classes.
C	Confusion set	Format does not decide the type; use does.
D	Camouflage	No peer comparison is involved.

2. Answer B Concept 2

A parent ~~at a school with a strong music program~~ reads that her son scored at the 70th percentile on a national reading test. ~~She has three children in the district.~~ She concludes he got 70% of the questions right. How should the teacher explain the score?

Target word: interpret a percentile **Decides:** Percentile compares to a norm group.

A	Camouflage	The norm group is not his family.
B	Correct	Percentiles are norm-referenced ranks.
C	Adjacent category	Proficiency is criterion-referenced.
D	Confusion set	Confuses percentile with percent correct.

3. Answer A Concept 2 · transfer

A state test reports each student as below, approaching, meeting, or exceeding the standard based on fixed cut scores. What type of interpretation is this?

Target word: reverse: classify from the report **Decides:** Comparison to fixed standards.

A	Correct	Scores are judged against set standards.
B	Adjacent category	Describes use, not score interpretation.
C	Inverted operation	Norm-referenced compares to peers.
D	Confusion set	Percentiles are norm-referenced.

4. Answer B Concept 3

A math test on solving word problems is written at a reading level three grades above the students' level. Students' scores are nearly identical when they retake it a week later. ~~The test was designed by a team of five teachers.~~ What is the main concern?

Target word: validity or reliability **Decides:** Reading level interferes with measuring math.

A	Confusion set	The stem says scores are consistent.
B	Correct	The test is consistent but may measure the wrong skill.
C	Inverted operation	Reliability does not guarantee validity.
D	Camouflage	Team size is irrelevant.

5. Answer D Concept 4

A second grader in a school with a new reading coach has weekly fluency probes. The aim line calls for gains of 2 words correct per minute each week. Her last four scores fell below the aim line. ~~She loves art class.~~ What should the team do?

Target word: decision rule **Decides:** Four consecutive points below the aim line.

A	Camouflage	Irrelevant.
B	Inverted operation	Raising the goal fits scores above the aim line.
C	Dropped element	Ignores the decision rule already met.
D	Correct	A consistent trend below the goal signals a change.

6. Answer C Concept 4 • transfer

A student read 40 words correct per minute in week 1 and 64 in week 9. What is the average weekly rate of improvement?

Target word: compute rate of improvement **Decides:** $(64 - 40) / (9 - 1)$ weeks.

A	Dropped element	Total gain, not per week.
B	Inverted operation	Divides the ending score by 9 ($64/9$) instead of dividing the gain by the interval.
C	Correct	$24 / 8 = 3$.
D	Wrong stem number	Divides by 9 instead of 8 intervals.

7. Answer C Concept 5

On a unit test given in a newly remodeled classroom, 30 students answered item 7. Only 9 answered correctly, and most of the top-scoring students chose the same wrong answer. ~~The test took 45 minutes.~~ What should the teacher conclude?

Target word: interpret item statistics **Decides:** Difficulty $9/30 = 0.30$ and high scorers missed it.

A	Camouflage	Timing is irrelevant.
B	Absolute language	Top students missing an item often points to the item, not them.
C	Correct	$9/30 = 0.30$; strong students missing it signals poor discrimination.
D	Confusion set	A low p-value means few answered correctly: a hard item.

8. Answer D Concept 5 • transfer

An item has a difficulty of 0.85 on a test taken by 40 students. How many students answered it correctly?

Target word: solve for a count **Decides:** Correct = 0.85×40 .

A	Wrong stem number	Reads the proportion as a count.
B	Confusion set	Divides 40 by 0.85.
C	Inverted operation	That is the number who missed it.
D	Correct	$0.85 \times 40 = 34$.

9. Answer A Concept 6

A teacher ~~at a school with a new theater~~ wants students to see exactly which parts of their argumentative essays need work: claim, evidence, reasoning, and conventions. ~~She teaches 110 students.~~ Which scoring tool fits best?

Target word: rubric type **Decides:** Feedback is needed on each criterion.

A	Correct	Shows strengths and needs per criterion.
B	Dropped element	No information on specific criteria.
C	Camouflage	Faster, but it does not give criterion-level feedback.
D	Adjacent category	Measures completion, not quality.

10. Answer D Concept 7

A student ~~in a school with a large library~~ turns in a lab report. The teacher, ~~who coaches volleyball~~, returns it three weeks later. Which feedback would be most effective if given promptly?

Target word: effective feedback **Decides:** Specific, goal-linked, and actionable.

A	Camouflage	Irrelevant and not about the work.
B	Dropped element	Grade and praise with no guidance.
C	Absolute language	Vague and not actionable.
D	Correct	Specific, tied to the goal, with a next step.

11. Answer C Concept 8

A teacher ~~at a school with a new front office~~ is presenting test results to families. Scores for ten students were 72, 75, 76, 78, 78, 80, 81, 82, 84, and 4 (a student who left after one question). Which statistic best represents the typical score?

Target word: choose the statistic **Decides:** One extreme low outlier.

A	Adjacent category	Range shows spread, not typical score.
B	Wrong stem number	An outlier, not typical.
C	Correct	Middle values are 78 and 78; the median resists the outlier.
D	Confusion set	The outlier drags the mean ($710/10 = 71$) below nearly every student.

12. Answer B Concept 8 · transfer

A class's test results show a mean of 74 and a median of 86. What does this pattern most likely mean?

Target word: reverse: infer the data from the statistics **Decides:** Mean well below median signals a few very low scores.

A	Confusion set	Treats the distorted mean as the typical score.
B	Correct	Low outliers lower the mean but barely move the median.
C	Absolute language	Mean and median differ whenever data are skewed.
D	Inverted operation	High outliers would put the mean above the median.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Formative vs. summative is about use	An assessment is formative when results are used to adjust teaching and learning, and summative when used to judge learning at the end of a period.	Classifying by format (quiz vs. test) instead of by use.	What the teacher does with the results.
2	Norm- vs. criterion-referenced	Norm-referenced scores compare a student to a peer group (percentiles); criterion-referenced scores compare to a fixed standard (proficient or not).	Reading a percentile as percent correct.	A percentile rank vs. a cut score.
3	Validity vs. reliability	Validity is whether scores support the intended interpretation; reliability is consistency of scores. A test can be reliable but not valid.	Assuming consistent scores are accurate for the purpose.	A test that measures something other than the skill.
4	Progress monitoring decisions	Compare a student's trend to the goal line; several data points below the aim line signal an instructional change.	Reacting to a single data point.	Weekly probe scores and a goal.
5	Item analysis	Item difficulty is the proportion who answered correctly; an item missed more by high scorers than low scorers has negative discrimination and should be reviewed.	Reading a low difficulty value as an easy item.	Counts of correct answers on an item.
6	Analytic vs. holistic rubrics	Analytic rubrics score each criterion separately and give detailed feedback; holistic rubrics give one overall score and are faster.	Using a holistic rubric when students need criterion-level feedback.	A goal of pinpointing strengths and weaknesses.
7	Effective feedback	Effective feedback is timely, specific to the goal, and tells students what to do next.	Giving only a grade or general praise.	Choosing the most useful comment on student work.
8	Mean vs. median when reporting	With outliers or skewed scores, the median represents the typical student better than the mean.	Reporting the mean when a few extreme scores distort it.	A data set with one or two extreme values.

Course scope (public description): Monitoring Student Learning Through Assessment enhances students' skills in evaluating learner progress and educational outcomes. This course provides an in-depth exploration of various assessment techniques, including formative and summative assessments, standardized tests, benchmark assessments, progress monitoring, and alternative assessment strategies. This course teaches students to design effective assessment tools, interpret data to inform instruction, and provide meaningful feedback to learners. This course also provides a foundation of data analysis that supports educators' need to understand data and present data to stakeholders. It also explores online and digital assessment tools. This course requires students to design an assessment based on evidence-based practices.